Module 5: Data Collection, Cleaning, Visualization, and Analysis

Theoretical Assignment 1: Importance of Data Cleaning in Data Science

Data cleaning is a critical step in every data science project. It involves detecting and correcting errors, removing inconsistencies, and handling missing or invalid data. Clean data ensures accurate, reliable, and meaningful analysis. Key steps in data cleaning: - Handling Missing Values - Removing Duplicates - Correcting Data Types - Handling Outliers - Standardization and Normalization Without proper cleaning, even the best models will fail because their foundation — the data — is flawed.

Theoretical Assignment 2: Data Visualization Techniques & Best Practices

Data visualization transforms data into graphical form for easier understanding and insight discovery. Common techniques: - Bar charts, Histograms - Scatter plots, Line graphs - Boxplots, Heatmaps, Pie charts Best practices: - Label axes clearly - Use appropriate chart types - Avoid clutter and distortion - Highlight key insights Visualization bridges the gap between complex data and human understanding.

Practical Task: Car Dataset Cleaning and Visualization

Dataset: https://www.kaggle.com/datasets/CooperUnion/cardataset

Steps performed: 1. Import libraries 2. Load dataset 3. Perform data cleaning (missing values, duplicates, data types, outliers) 4. Feature engineering 5. Visualization and analysis 6. Save cleaned dataset

1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid", palette="Set2")
```

2. Load Dataset

```
df = pd.read_csv("cars.csv")
print(df.shape)
df.head()
```

3. Data Overview

```
df.info()
df.describe()
df.isnull().sum().sort_values(ascending=False)
df.duplicated().sum()
```

4. Handle Missing Values

```
df = df.drop_duplicates()
num_cols = df.select_dtypes(include=['float64', 'int64']).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
cat_cols = df.select_dtypes(include=['object']).columns
for col in cat_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)
```

5. Data Type Correction

```
df['Year'] = df['Year'].astype(int)
df['Make'] = df['Make'].astype('category')
```

6. Outlier Detection and Removal

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['MSRP'])
Q1 = df['MSRP'].quantile(0.25)
Q3 = df['MSRP'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['MSRP'] >= Q1 - 1.5*IQR) & (df['MSRP'] <= Q3 + 1.5*IQR)]
```

7. Feature Engineering

```
df['Avg_MPG'] = (df['City MPG'] + df['Highway MPG']) / 2
```

8. Visualization

```
# Top 10 car makes by average price
plt.figure(figsize=(10,6))
top_makes = df.groupby('Make')['MSRP'].mean().sort_values(ascending=False)[:10]
sns.barplot(x=top_makes.values, y=top_makes.index)
plt.title("Top 10 Car Makes by Average MSRP")

# Correlation heatmap
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")

# Horsepower vs Price
sns.scatterplot(x='Horsepower', y='MSRP', hue='Make', data=df, alpha=0.6)
plt.title("Horsepower vs. Price by Brand")
```

9. Save Cleaned Dataset

```
df.to_csv("cleaned_car_data.csv", index=False)
print("■ Cleaned dataset saved successfully!")
```

Insights:

- Luxury brands like BMW and Mercedes-Benz have the highest average MSRP. - Horsepower and Engine Size show strong correlation with Price. - Higher fuel efficiency (MPG) tends to reduce MSRP.

Deliverables:

1. Essay: Importance of Data Cleaning in Data Science 2. Presentation: Data Visualization Techniques & Best Practices 3. Cleaned Dataset: cleaned_car_data.csv 4. Jupyter Notebook: Module5_CarData_Cleaning_Visualization.ipynb